1. PROBLEM STATEMENT

Forests are increasingly afflicted by drought, pests, and diseases, raising concern as they remain an essential part of the environment. They provide critical natural habitats, resources, and recreational areas, while absorbing enormous amounts of carbon dioxide. Lab testing and manual inspection, which are methods currently used to determine a tree's health, are time consuming and often commissioned only after visible damage is observed or require cutting the tree. A solution; Forest Buddy is needed to address this problem that provides a low-cost, reliable, and non-intrusive method.

1.1. Need Statement

According to the U.S. Department of Agriculture (USDA), nearly 11-16% of tree species are being driven to extinction by insects and disease, while 11% of forests in the United States are exposed to drought, expected to rise to 38% by mid-century [1], [2]. Low water content can indicate that a tree is suffering from a disease such as vascular wilt or root rot, but it could also simply mean a healthy tree is experiencing drought stress. Current methods are often time-consuming or require visible indicators of tree decay, often making the necessary interventions too late for any individual tree or potentially an entire surrounding population [3]. A portable, reliable, and non-intrusive device for monitoring tree health is needed to allow forest health to be assessed more efficiently.

1.2. Objective Statement

The objective of this project is to design and build an autonomous rover that monitors the health of trees and forests. Forest Buddy measures changes in GPS signal attenuation as the signal passes through tree branches and leaves. Since the amount of water in vegetation affects GPS signal strength, the signal attenuation can be used to estimate the water content in the tree. Forest Buddy is guided by user-defined waypoints set by the operator walking through the forest and can be remotely controlled if necessary. As the rover travels through the forest, it collects GPS data and transmits this data to a database for analysis. The collected data is compared with control data from a base station which is in an open field.

1.3. Background and Related Work

Currently, forest health is assessed using manual inspection, drone networks, and fixed sensor networks. These methods primarily rely on visible symptoms, such as needle discoloration, or directly detecting a pathogen, rather than measuring water content [4], [5]. Manual inspection of forests is impractical for large areas [4], [5]. Remote sensing techniques such as drone and satellite imaging can capture broader spectral data like chlorophyll levels and canopy color, but environmental conditions and coverage limitations can make consistent, large-scale monitoring difficult [4], [5]. Fixed sensor networks can continuously monitor tree health, but require installing a device on each tree, which is expensive for large areas [4]. Another promising device is Leaf Link, a former capstone project using a pulley system under a canopy. Forest Buddy mobilizes tree population diagnostics by putting these capabilities on a rover, with a focus on large area coverage. Forest Buddy will be effective at detecting sickness in a time efficient, inexpensive, non-intrusive, and scalable way by using water content as an early indicator of sickness. Mobility allows it to collect more data from a larger number of trees, helping researchers monitor forests more efficiently.

2. DESIGN REQUIREMENT SPECIFICATIONS

Forest Buddy measures how much a tree canopy weakens GPS signals, converts that loss into a health estimate, and delivers the result to researchers without harming the tree. Forest Buddy's design specifications define the expectations the rover must meet to serve its intended users in a forest environment. These requirements reflect the project's goals, expected operating conditions, and available resources. They provide a basis for making design decisions and determining whether the completed rover performs as intended. The following sections describe the required capabilities, establish measurable performance targets, and explain why each requirement is important to the system's overall purpose.

2.1. Features and Requirements

Forest Buddy's primary features and requirements focus on helping the rover follow an operator safely and reliably through a forest environment. Accurate location and orientation information are needed to maintain the rover's position and direction during travel. The system also requires dependable power and sufficient operating time to cover the intended area without frequent interruptions. Forest terrain can include slopes and obstacles; the rover must maintain a suitable following distance, detect obstructions, and move safely over uneven ground.

2.1.1. Features and System Design Requirements

Table 2-1 outlines the main system design requirements for Forest Buddy based on its core functions. These requirements set measurable performance goals to help the system operate efficiently and reliably while meeting user needs under expected operating conditions.

Table 21: Features and System Design Requirements

Features	System Design Requirements	Justification
1	The rover will maintain operational status for at least 60 minutes. Operational status is defined as being able to drive around and take samples as well as when it is stationary.	This requirement is based on the expected time a user will operate the device.
2	Computer vision must detect human subjects upon entering frame within 2 seconds.	The camera must track the person in the frame quickly to allow ease of operation.
3	The IMU measurements are within $\pm$ 5 degree of the rover's true orientation.	This requirement is necessary for maintaining proper orientation of the rover, both for navigation and for accurate GNSS readings.
4	The emergency stop must stop the rover 1 to 3 feet from a detected obstacle within 1 second. While it starts to ramp down once a object is detected at 5 feet.	This prevents the rover from harming people, trees, or itself by flipping or falling over from hitting an object.
5	The rover must be able to climb and descend an inclined 20% grade plane made of gravel or dirt.	The rover must be able to navigate rolling hills, gravel, and dirt.

Features:
1. The system must run for at least an hour, covering a large area with regulated power.
2. The system has precise and reliable human detection that can be done in a timely manner.
3. The system has precise localization and can maintain proper orientation.
4. The system does not run into obstacles and maintains proper following distance from the operator.
5. The system must be able to travel across terrain that is expected in a forest including stumps, exposed tree roots, overhanging branches and vines, mild slopes, mud, leaf litter, and debris.

The rover must maintain operations for at least 60 minutes. This requirement is based on the expected duration of an operation for Forest Buddy to collect sufficient data during a single deployment. Maintaining this operating time ensures that the rover does not need to be frequently stopped to recharge or change the battery. The computer vision on the rover must be able to detect a human subject within 2 seconds upon entering the camera's view. This allows the operator to quickly begin using the rover without requiring extended setup time. The camera will also track the operator as they travel through the forest. Reliable human tracking is important as it is the primary way to establish the waypoints that the rover will be following.

The IMU measurements must remain within 1 degree of the rover's true orientation. Accurate orientation is necessary for determining the rover's heading and movement. It also helps maintain an accurate position when combined with GNSS measurements. The emergency stop feature must stop the rover between 3 to 5 feet from a detected object. This safety feature prevents the rover from colliding with objects such as trees or people. The rover must also be able to climb inclines of a 20% grade. This requirement allows Forest Buddy to operate on moderately uneven terrain as expected in a forest.

2.1.2. Subsystem Requirements and Demonstration Setup

Table 2-2 introduces the subsystem requirements for Forest Buddy while also giving the demonstration setups. The requirements must be proven using a measurable test which is provided in the table below.

Table 222: Subsystem Requirements and Demonstration Setup

Subsystem/ Item:	Requirements	Demonstration Setup
Camera Vision	(1) The subsystem must detect and identify the first operator that comes into the camera view so the rover can follow them. It should identify them within 2 seconds. (2) The subsystem must track the detected operator as they move through the forest. If the operator was to become fully obstructed and the system lost them the rover will stop and require them within 2 seconds of them becoming visible again. Visibility is determined if at least 40-50% of their body is showing. (3) The subsystem must detect obstacles that are greater than 5” tall in its path and alert the operator.	(1) A person will be placed in front of the rover, and the camera will detect and identify them as the operator. The system will then have an indicator that the operator has been detected. A second person will then walk into frame; the system will remain tracking of the first person. (2) The operator will walk through a simulated forest-like test area while the rover’s camera is tracking them. The system will be continuously updating and providing the person’s position to the navigation subsystem. The operator will also walk behind objects to test the system's ability to maintain or regain tracking. (3) Object such as sticks and stones ranging from 2” to 10” will be placed in front of the rover’s path. It will then be tested that as the rover approaches one of these objects, it should stop and alert the operator if it cannot go over it.
Locomotion	(1) The subsystem must be able to climb a 20% graded slope. (2) The subsystem must be able to travel over obstacles less than 3” tall. (3) The subsystem must be able to stop the rover from full speed to a full stop within 1 second.	(1) Forest Buddy will be placed at the bottom of a wedge at an angle of approximately 12 degrees and attempt to drive up and down the incline. (2) Several obstacles smaller than 3” will be placed directly in front of Forest Buddy’s wheels, which will then attempt to drive over them. (3) Forest Buddy will be driven forward at maximum speed, then the E-Stop will be engaged. The rover must stop within 1 second of the E-Stop button being engaged.

Data Analysis	(1) The subsystem must display the collected data as a heat map on the Forest Buddy website, using a color scale with 5 labeled levels from low to high attenuation, and a legend that shows what each color means. (2) The subsystem must display the collected data as a line graph on the Forest Buddy website, with attenuation (dB-Hz) on the vertical axis and time on the horizontal axis, and both axes labeled. (3) The subsystem must load the Forest Buddy webpage with the collected data displayed within 3 seconds of opening the page.	(1) The Forest Buddy website is opened with the collected data from the road run. The heat map is checked for a color scale with 5 labeled levels from low to high attenuation, and for a legend showing what each color means. The presence of the heat map, all 5 levels, and the legend is verified. (2) The Forest Buddy website is opened with the same data. The line graph is checked for attenuation in dB-Hz on the vertical axis and time on the horizontal axis, with both axis labeled. The presence of the line graph and both axis labels is verified. (3) A screen recording is started, and the Forest Buddy webpage is opened with the collected data. The time from opening the page until the data appears is read from the recording. This is repeated for 3 loads, and a time of 3 seconds or less for every load is verified.
Communication	(1) The subsystem must maintain a wireless connection between Forest Buddy and the base station at 200-300 ft. If signal is disconnected, the remaining signals that need to be sent will buffer until connection is reestablished. (2) The subsystem must have positional accuracy at the centimeter level. (3) The subsystem must allow the user to control the locomotion of Forest Buddy through a remote controller. The response time from the controller to robot needs to be less than 2 seconds at less than 200' away.	(1) Forest Buddy will be placed next to the base station to establish a wireless connection. Then Forest Buddy will be moved away from the base station approximately 200 ft. The data will be transmitted between Forest Buddy and the base station while the wireless connection is maintained. The user will look at the received messages at the base station to confirm the data is being sent successfully and the connection is maintained. If the signal is disconnected during this time, the user will have to look at the time stamps provided on Forest Buddy in order to make sure the signals were buffered correctly. (2) Forest Buddy will be placed at a fixed location, and its GNSS position will be recorded without communication to the base station. Without moving Forest Buddy, communication with the base station will then be established, and the GNSS position will be recorded again. The two sets of position data will be compared to verify that the system operates with centimeter-level positional precision. (3) To demonstrate this test, Forest Buddy is moved into an open area at least 200 ft

		from the base station. The user will then use the remote control to command the movement. The user will show the remote controller can move the rover forward, backward, left, and right. After it is confirmed the remote controller moves Forest Buddy, the response time will be recorded. A stopwatch will be used and the forward command will be given through the controller. Once given, Forest Buddy has less than 2 seconds to begin moving forward.
Navigation/ Localization	(1) The subsystem must measure the rover's orientation within ± 5 degree of its true orientation. (2) The subsystem must allow Forest Buddy to follow the operator during initial setup between 5 and 10 feet behind the operator and save the traveled path for future use. (3) The subsystem must use a navigation algorithm to guide Forest Buddy through saved waypoints in the correct order, reaching within 2 feet of each waypoint without requiring the operator to always lead the rover.	(1) Forest Buddy will be placed on a level surface, and its IMU orientation readings will be recorded. The rover will then be tilted forward, backward, left, and right. At each position it will be measured to compare to the level surface to verify the difference is no greater than ± 5 degree. (2) The operator will walk a marked path with straight sections and turns while the Forest Buddy follows. The rover will be observed to confirm that it responds to the operator's movements and stays within the specified range. After setup, the stored route will be reviewed to confirm that the traveled path was saved. (3) A predefined set of waypoints will be loaded into Forest Buddy. The rover will begin at the designated starting point and navigate the route while the operator remains stationary. Its recorded positions will be checked to verify that it reaches the specified distance while also making sure it reaches each waypoint in the correct order.
Signal Processing	(1) The subsystem must log data at the rate of between 8 and 15 Hz, feeding this stream of recorded data back to the base station. (2) The subsystem must measure barometric pressure, humidity, and temperature to benchmark receiver measurements. Calculated altitude must be within $\pm$ 1 cm of the GNSS receiver's found altitude. (3) The subsystem must record the GNSS message and sensor data as a, encoded CSV file, with a timestamped datapoint for every 1 millisecond of operation.	(1) Serial communication between the rover and base station will be established. Then, the rate of logging will be evaluated at base station using the implemented software from Data Analysis. (2) To evaluate the accuracy of the measured barometric pressure and environmental sensors, the subsystem will compare the rover's GPS receiver's altitude and barometer altitude. Through software, the subsystem will

		evaluate how closely the values match to confirm what is estimated from the microcontroller within $\pm$ 1 cm of accuracy. (3) The rover will split data between the onboard microprocessor and the base station. The sensor fusion algorithm will interpret the onboard GNSS data, then encode and timestamp the data generated.

2.2. Constraints

Table 2-3 shows constraints that are imposed on Forest Buddy. The constraints describe the limitations that Forrest Buddy must operate in. These include most environmental constraints while also showing functionality as well. The limitations affect decisions for components, how each subsystem will work, and how the team will design Forest Buddy.

Table 2-3: Constraints

Type	Name	Description
Environmental	Weather	The Forest Buddy must operate in windspeeds of 10 mph and rainfall rates of 1mm of rainfall per minute per 10 minutes when tilted up to 15°.
Environmental	Terrain	Forest Buddy must operate in forested environments with a 20% grade and maneuver in broken ground of up to 2” in height.
Environmental	Obstacles	Forest Buddy must be able to drive over small debris typical of a forest environment. Including rocks, roots, sticks, limbs, leaves of up to 5” in height.
Safety	E-Stop	The rover must have a physical E- Stop button that stops all motor movement within 1 second of being pressed.
Functionality	Width	Forested lands often include trees that are 5’ – 10’ apart so the rover must be able to maneuver in between these trees.

Forest Buddy is a semi-autonomous and non-intrusive forest ranging rover that operates outside to collect the proper data to determine forestry and tree health. The environmental constraints listed in the table force the rover to handle most outside factors. Knowing these limitations that are imposed on Forest Buddy, the team can focus on designs to handle these constraints.

2.3. Standards

Forest Buddy operates under IP-52, meeting the standard for housing its components in a dust-tight environment, where tested particles cannot interfere with the operation of the rover, alongside being protected from light splashes of water it may encounter in forested terrain. Forest Buddy additionally may become a hazard for people in the path of its route in the event of maloperation, and as such, features two methods of emergency stop; An autonomous stop when an obstacle is detected, and an accessible emergency stop button. Forest Buddy's receivers additionally maintain consistent positioning in accordance with ISO 17123, with RTK allowing the receivers to be within ± 1 cm of true location. Table 2-4 includes specific benchmarking that corresponds to a device meeting these standards, alongside the document sources of each mentioned standard.

Table 2-4: Engineering Standards

Specific Standard	Standard Document	Specification / Application
IP-52	The product meets Ingress Protection Standard 60529 set by the International Electrotechnical Commission [6].	Forest Buddy is dust-protected from 1 kg of <_100 micrometer particles for up to 1 hour and protected from up to 3 mm of water per minute for up to 10 minutes.
ISO 13850	The product follows International Organization for Standardization standard 13850 [7].	Forest Buddy features a physical emergency stop button that initializes a category 1 controlled stop, with restart requiring manual or teleoperated control.
ISO 17123-8:2015	Internation Organization for Standardization standard 17123 [8].	Forest Buddy's rover and base station maintain proper position within 5% error of the manufacturer's claims of precision, and ± 1cm accuracy of claimed position for both horizontal distance and ±1.5 cm for claimed vertical position.

The Ingress Protection Standard 60529, IP-XX series contain specifications for the amount of dust and water allowable in the enclosure surrounding the components. The first number indicates how much dust is allowable and the extent to which the particles can interfere with the rover's operation, while the second number indicates how much water the rover can tolerate. For Forest Buddy, 5 indicates the rover's operation cannot be interfered with by dust particles, although particles <100 micrometers can be in the enclosure, while 2 indicates the rover can maintain operation in the event of "light splashes", defined as protected from up to 3 mm of water for up to 10 minutes. ISO 13850 deals with the standards surrounding the operation of emergency stops. The manually operated emergency stop is red and easily accessible. Forest Buddy will feature a category "1" emergency stop, where movement is stopped instantly on operation, and power is

maintained for just a few seconds to bring the device to a controlled halt. ISO 17123 deals with the standards surrounding the operation of RTK devices. Forest Buddy's operation is within the required precision of the true position as calculated by the GNSS receiver module.

2.4 REFERENCES

[1] “Forests and a future of drought: Area exposed to severe drought projected to triple by 2070 | US Forest Service Research and Development,” US Forest Service Research and Development, 2026. https://research.fs.usda.gov/srs/articles/forests-and-future-drought-area-exposed-severe-drought-projected-triple-2070 (accessed Sept. 03, 2026).

[2] “Researchers complete first comprehensive threat assessment of all U.S. trees,” *United States Botanic Garden*, Aug. 23, 2022. https://www.usbg.gov/blog/researchers-complete-first-comprehensive-threat-assessment-all-us-trees (accessed Sept. 08, 2026).

[3] “FIA FAQ - Colorado State Forest Service,” *Colorado State Forest Service*, Jan. 27, 2017. https://csfs.colostate.edu/science/forest-inventory-analysis/fia-faq/ (accessed Sept. 14, 2026).

[4] M. Li, H. Li, X. Ding, L. Wang, X. Wang, and F. Chen, "The detection of pine wilt disease: A literature review," Int. J. Mol. Sci., vol. 23, no. 18, Art. no. 10797, Sep. 2022, doi: 10.3390/ijms231810797. [Online]. Available: https://doi.org/10.3390/ijms231810797 (accessed Sept. 8, 2026).

[5] S. Tahir, S. S. Hassan, L. Yang, M. Ma, and C. Li, "Detection methods for pine wilt disease: A comprehensive review," Plants, vol. 13, no. 20, Art. no. 2876, Oct. 2024, doi: 10.3390/plants13202876. [Online]. Available: https://doi.org/10.3390/plants13202876 (accessed Sept. 8, 2026).

[6] IEC, *Degrees of protection provided by enclosures — IP code*, IEC 60529, 20th ed., Geneva: International Electrotechnical Commission, 2020. (Accessed 6 Oct. 2026).

[7] ISO. ISO 13850:2015 Safety of machinery — Emergency stop function — Principles for design. ISO.org. (Accessed 6 Oct. 2026).

[8] International Organization for Standardization/International Electrotechnical Commission, “ISO/IEC 17123 — Calibration and verification of measurement devices,” ISO/IEC 17123, 1st ed., 2023, pp. 1–120. Available: https://www.iso.org/standard/62085.html. (Accessed: Oct. 6, 2026)

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